

PE-EC801B — Fiber Optic Communication

Semester VIII • 3L:0T:0P • 3 Credits

Module 3 Exam Notes

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Optical Sources — LED

Light-Emitting Diode (LED)

An LED is a forward-biased p-n junction semiconductor device that emits light by **spontaneous emission** when electrons recombine with holes. LEDs for fiber optics operate in the 850 nm, 1310 nm, or 1550 nm windows.

1.1 Surface-Emitting vs Edge-Emitting LED

- **Surface-Emitting LED (SLED / Burrus LED):** Emits light perpendicular to the junction plane. Large emitting area ($\sim 50 \mu\text{m}$ diameter), Lambertian emission pattern, easy to couple to MMSI/MMGI fibers. Modulation bandwidth $\sim 20\text{--}100$ MHz.
- **Edge-Emitting LED (ELED):** Emits light parallel to the junction from the cleaved edge of the active stripe. Narrower emission cone, better coupling to single-mode or small-core multimode fibers. Modulation bandwidth $\sim 100\text{--}400$ MHz. Output spectrum FWHM $\sim 30\text{--}80$ nm.

LED Output and Modulation Bandwidth

$$P_{opt} \propto I_d \cdot \eta_{ext}, \quad P(\omega) = \frac{P(0)}{\sqrt{1 + (\omega\tau_r)^2}}$$

Modulation bandwidth: $f_{3dB} = \frac{\sqrt{3}}{2\pi\tau_r}$, where τ_r is the minority carrier radiative lifetime ($\sim 2\text{--}10$ ns).

Advantages of LEDs: Simple construction, lower cost, longer lifetime ($> 10^7$ hours), linear L-I characteristic, no threshold, operable without temperature control, immune to catastrophic failure.

Disadvantages: Broad output spectrum (large material dispersion), lower output power, slower modulation compared to laser diodes.

2. Optical Sources — Laser Diodes

Laser Diode

A **laser diode** (LD) is a semiconductor p-n junction in which **stimulated emission** dominates above a threshold current I_{th} . It produces coherent, narrow-spectrum, high-power output suitable for SMF and high-speed links.

2.1 Fabry-Perot (FP) Laser

Uses two cleaved facets as reflective mirrors forming a Fabry-Perot cavity. Oscillation occurs at wavelengths satisfying the resonance condition $L = m\lambda/(2n)$ (m integer). FP lasers are **multi-longitudinal-mode**; spectrum is a comb of modes separated by $\Delta\lambda = \lambda^2/(2nL)$.

2.2 DFB (Distributed Feedback) Laser

A Bragg grating is integrated into the active layer. Feedback is distributed along the cavity. Only the mode satisfying the Bragg condition $\lambda_B = 2n\Lambda$ (where Λ is the grating period) is amplified. DFB lasers are **single-longitudinal-mode**, essential for WDM systems.

2.3 Rate Equations and Threshold Current

Laser Rate Equations and L-I Characteristic

Carrier density N and photon density S :

$$\frac{dN}{dt} = \frac{\eta_i I}{eV} - \frac{N}{\tau_n} - Gg_0(N - N_0)S$$

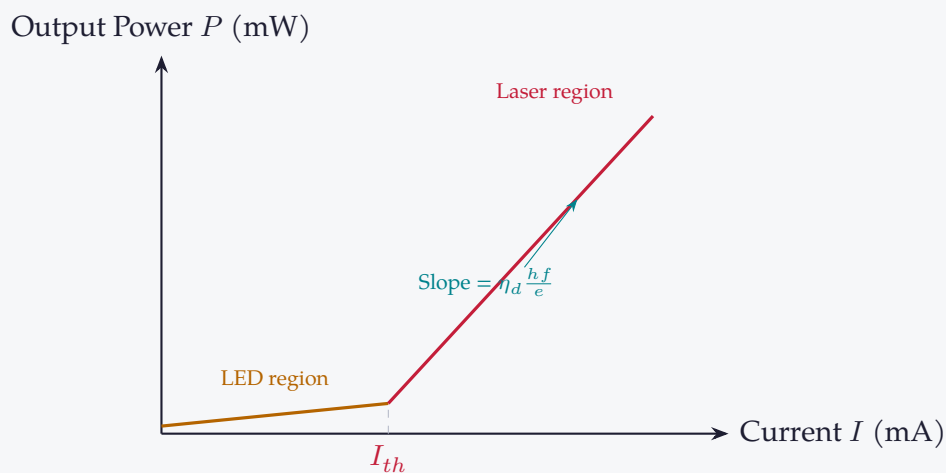
$$\frac{dS}{dt} = Gg_0(N - N_0)S + \frac{\beta N}{\tau_n} - \frac{S}{\tau_p}$$

Threshold current: I_{th} is where gain = loss (photon density starts growing exponentially).

$$P_{out} = \eta_d \cdot \frac{hf}{e}(I - I_{th}), \quad I > I_{th}$$

where η_d is differential quantum efficiency, hf is photon energy.

L-I Curve of a Laser Diode



3. Photodetectors — PIN Diode and APD

Photodetector — Responsivity and Quantum Efficiency

A **photodetector** converts optical power into electrical current via the photoelectric effect. **Responsivity** $\mathcal{R}$ (A/W) is the photocurrent per unit optical power. **Quantum efficiency** η is the fraction of incident photons that generate electron-hole pairs.

Responsivity Formula

$$\mathcal{R} = \frac{I_p}{P_{opt}} = \frac{\eta e \lambda}{hc} \quad (\text{A/W})$$

where η = quantum efficiency, e = electron charge (1.6×10^{-19} C), λ = wavelength (m), h = Planck's constant (6.626×10^{-34} J·s), c = speed of light.

Also: $\mathcal{R} = \eta e / (hf)$ where $f = c/\lambda$ is optical frequency.

3.1 PIN Photodiode

A **PIN diode** has a lightly doped intrinsic (I) layer between heavily doped P and N regions. The wide depletion region extends across the I layer under reverse bias, allowing efficient absorption of photons over a large volume.

- **Structure:** $P^+ / I(\text{intrinsic}) / N^+$, reverse-biased.
- **Bandwidth:** Limited by transit time $\tau_{tr} = d/v_d$ and RC time constant. $f_{3dB} \approx 1/(2\pi\tau_{tr})$. Bandwidths up to tens of GHz are achievable.
- **Responsivity:** Typically $\mathcal{R} \approx 0.6\text{--}0.9$ A/W at 1300–1550 nm (InGaAs PIN).

3.2 Avalanche Photodiode (APD)

An APD operates under high reverse bias where photogenerated carriers gain sufficient energy to ionize more carriers through **impact ionization** — the avalanche effect. This provides internal current gain M .

APD Gain, Noise, and SNR

APD multiplication gain: $M = I_{APD} / (I_{primary})$.

Excess noise factor: $F(M) \approx k_A M + (1 - k_A)(2 - 1/M)$ (McIntyre approximation), where k_A is the ionization ratio ($k_A = \alpha_h / \alpha_e$).

For InGaAs APD: M typically 10–30; $k_A \approx 0.4\text{--}0.7$.

3.3 Noise in Photodetectors

- **Shot noise:** Due to the quantum nature of photon detection. $\langle i_s^2 \rangle = 2eI_p B$ (PIN); $\langle i_s^2 \rangle = 2eI_p M^2 F(M) B$ (APD), where B is bandwidth.
- **Thermal noise (Johnson noise):** Due to random thermal agitation of electrons in the load resistor R_L . $\langle i_T^2 \rangle = 4k_B T B / R_L$, where $k_B = 1.38 \times 10^{-23}$ J/K.
- **SNR:**

$$\text{SNR}_{PIN} = \frac{(\mathcal{R}P_{opt})^2}{2e\mathcal{R}P_{opt}B + 4k_B T B / R_L}$$

$$\text{SNR}_{APD} = \frac{(M\mathcal{R}P_{opt})^2}{2eM^2 F(M)\mathcal{R}P_{opt}B + 4k_B T B / R_L}$$

3.4 Optical Receivers — Pre-Amplifier Types

- **High-impedance (HZ) pre-amplifier:** Large R_L minimizes thermal noise, but limits bandwidth due to RC roll-off. Equalizer needed. Best sensitivity.

- **Transimpedance (TIA) pre-amplifier:** Uses feedback resistor R_f to convert photocurrent to voltage. Flat frequency response up to higher bandwidth with good sensitivity. Preferred in practice.

4. Optical Link Design — BER, Q-Factor, and Receiver Sensitivity

Optical link design is the process of ensuring that a fiber optic system meets both the **power budget** (sufficient received power for the target BER after accounting for all losses) and the **dispersion budget** (pulse broadening stays within the allowed bit-period fraction). The power budget gives: $P_{tx} - (\text{total losses}) \geq P_{rx,min}$, where losses include fiber attenuation (αL), splice losses, connector losses, and a system margin ($\sim 3\text{--}6$ dB). The dispersion budget requires: $\sigma_{total} \leq T_b/(4\sqrt{2})$ (Gaussian pulse limit), where $\sigma_{total} = \sqrt{\sigma_{modal}^2 + \sigma_{chromatic}^2 + \sigma_{source}^2}$.

BER and Q-Factor

Bit Error Rate (BER) is the probability that a received bit is in error. The **Q-factor** is the signal-to-noise amplitude ratio that directly maps to BER. For a Gaussian noise model: $\text{BER} = \frac{1}{2}\text{erfc}(Q/\sqrt{2})$.

Link Design Equations

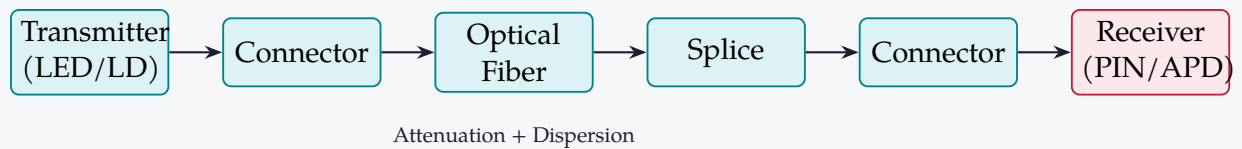
Q-factor	$Q = \frac{I_1 - I_0}{\sigma_1 + \sigma_0}$ (I_1, I_0 : mean currents for 1, 0)
BER vs Q	$\text{BER} \approx \frac{1}{2}\text{erfc}\left(\frac{Q}{\sqrt{2}}\right) \approx \frac{\exp(-Q^2/2)}{Q\sqrt{2\pi}}$
BER = 10^{-9} requires	$Q \approx 6$
Quantum limit	$\text{BER}_{QL} = \frac{1}{2}e^{-\eta N_p}$, minimum $N_p \approx 10$ photons/bit
Receiver sensitivity	Minimum P_{min} (dBm) for target BER

4.1 Power Penalties

Power penalties are additional receiver power margins required due to system imperfections beyond the AWGN model.

Penalty Source	Cause	Typical Range
Modal noise	Mode partition noise in MMF; speckle pattern changes	0.5–1 dB
Extinction ratio	Non-zero optical power for “0” bit ($r = P_0/P_1 \neq 0$)	1–2 dB (for $r = 0.1$)
Timing jitter	Random deviation in bit timing $\Rightarrow$ ISI	0.5–2 dB
Chromatic dispersion	Pulse broadening causes ISI at high bit rate	Depends on $D \cdot B \cdot L$
Reflection noise	Multiple reflections create interferometric noise	0.5–1 dB

Block Diagram — Complete Fiber Optic Link



Quick Revision — Module 3

Optical Sources, Detectors, and Link Design | PE-EC801B

Key Formulas at a Glance

Responsivity	$\mathcal{R} = \eta e \lambda / (hc) \quad (\text{A/W})$
Shot noise (PIN)	$\langle i_s^2 \rangle = 2e\mathcal{R}P_{opt}B$
Thermal noise	$\langle i_T^2 \rangle = 4k_B T B / R_L$
LED modulation BW	$f_{3dB} = \sqrt{3} / (2\pi\tau_r)$
Laser output power	$P_{out} = \eta_d (hf/e)(I - I_{th}) \quad (I > I_{th})$
Q-factor	$Q = (I_1 - I_0) / (\sigma_1 + \sigma_0)$
BER (Gaussian)	$\text{BER} = \frac{1}{2} \text{erfc}(Q/\sqrt{2})$
BER = 10^{-9}	$Q \approx 6$
APD SNR	$\propto M^2 / F(M) \quad (\text{optimum } M \text{ exists})$

One-Line Definitions

- **Spontaneous emission:** Random photon emission by an excited electron without external stimulus (LED).
- **Stimulated emission:** A passing photon triggers an identical photon from an excited state (Laser).
- **Threshold current I_{th} :** The bias current above which laser gain exceeds cavity loss.
- **DFB laser:** Distributed Feedback laser; single-mode due to integrated Bragg grating.
- **Responsivity $\mathcal{R}$:** Photocurrent generated per watt of incident optical power (A/W).
- **Quantum efficiency η :** Fraction of photons that create an electron-hole pair.
- **APD gain M :** Internal multiplication of photocurrent by impact ionization.
- **Q-factor:** SNR amplitude ratio; $Q = 6$ gives $\text{BER} = 10^{-9}$.
- **Transimpedance amp:** Pre-amplifier with feedback R_f giving flat bandwidth and good sensitivity.

Frequently Asked / Viva Questions

Q1. What is the difference between an LED and a Laser diode?

LED uses spontaneous emission; broad spectrum; no threshold; lower power. Laser diode uses stimulated emission above I_{th} ; narrow coherent spectrum; higher power; higher modulation speed.

Frequently Asked / Viva Questions (contd.)

Q2. What is the threshold current in a laser diode?

It is the minimum bias current at which the gain of the active medium equals the total cavity losses, causing the onset of lasing (stimulated emission dominates).

Q3. Why is a DFB laser preferred for WDM systems?

A DFB laser emits a single longitudinal mode (due to the Bragg grating), giving a very narrow linewidth (<1 MHz). This is essential to prevent channel overlap in dense WDM systems.

Q4. Define responsivity and derive the formula $\mathcal{R} = \eta e \lambda / hc$.

$\mathcal{R} = I_p / P_{opt}$. Each photon of energy $hf = hc/\lambda$ generates η electron-hole pairs, contributing charge ηe . Per second, $P_{opt}/(hf)$ photons arrive, giving $I_p = \eta e P_{opt}/(hf) = \eta e \lambda P_{opt}/(hc)$, so $\mathcal{R} = \eta e \lambda / (hc)$.

Q5. What is the advantage of APD over PIN diode?

APD provides internal current gain M through avalanche multiplication, improving receiver sensitivity (lower minimum detectable power). However, it introduces excess noise $F(M)$ and requires higher bias voltage.

Q6. What is shot noise? When is it dominant?

Shot noise arises from the discrete quantum nature of photon detection. $\langle i_s^2 \rangle = 2eI_p B$. It dominates over thermal noise at high optical power or when the load resistor is large.

Q7. Define Q-factor and state its relation to BER.

$Q = (I_1 - I_0)/(\sigma_1 + \sigma_0)$. For Gaussian noise: $\text{BER} = \frac{1}{2} \text{erfc}(Q/\sqrt{2})$. $\text{BER} = 10^{-9}$ requires $Q \approx 6$.

Q8. What is the quantum limit of detection?

The absolute minimum BER is set by the Poisson statistics of photon arrival: $\text{BER} = \frac{1}{2} e^{-\eta N_p}$. The quantum limit is ~ 10 photons/bit for ideal detection.

Q9. What is extinction ratio penalty?

When the laser emits non-zero power for a "0" bit (extinction ratio $r = P_0/P_1 \neq 0$), the eye opening reduces, requiring extra received power to maintain the same BER. $\text{Penalty} = 10 \log_{10} \left(\frac{1+r}{1-r} \right)$.

Q10. Compare surface-emitting and edge-emitting LEDs.

SLED: wide emitting area, Lambertian pattern, easy coupling to MMF, BW ~ 100 MHz. ELED: narrow emission cone, better coupling to SMF or small MMF, higher BW ~ 400 MHz, higher power.

Q11. What is the difference between high-impedance and transimpedance pre-amplifiers?

HZ: large R_L minimizes thermal noise; limited bandwidth needing equalization. TIA: uses feedback R_f ; flat bandwidth; moderate noise; preferred for practical high-speed receivers.

LED vs Laser Diode — Comparison

Parameter	LED	Laser Diode
Emission type	Spontaneous	Stimulated
Coherence	Incoherent	Coherent
Spectral width	30–80 nm	<1 nm (DFB)
Threshold	None	I_{th} required
Modulation BW	~100–400 MHz	>10 GHz
Output power	Low (mW)	High (>10 mW)
Temperature sensitivity	Low	High
Cost	Low	High
Application	Short-haul, MMF	Long-haul, SMF, WDM